10,0

Avaliação Parcial 1

Curso: Tecnologia em Telemática

Disciplina: Cálculo Diferencial e Integral 1

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4,0 01. Determine o valor do seguinte limite:

$$\lim_{x\to 2} \frac{\sqrt{6x^2-5x+2}-4}{x^3-3x^2-7x+18} \qquad \frac{19}{-56}$$

3,0 02. Determine o valor do seguinte limite:

$$\lim_{x\to +\infty} \left(\frac{5x-17}{5x+11}\right)^{\frac{-15x}{7}} \qquad e^{12}$$

3,0 03. Calcule o valor do seguinte limite:

$$\lim_{x\to 0} \frac{1-Sec6x}{x^2} \qquad -18$$

$$\lim_{x\to 2} \frac{\sqrt{6x^2-5x+2}-4}{x^3-3x^2-7x+18} =$$

$$\lim_{x\to 2} \frac{\left(\sqrt{6x^2-5x+2}-4\right)\cdot h(x)}{\left(x^3-3x^2-7x+18\right)\cdot h(x)} =$$

Como $h(x) = \sqrt{6x^2-5x+2}+4$

$$\lim_{x\to 2} \frac{6x^2-5x-14}{\left(x^3-3x^2-7x+18\right)\cdot h(x)} =$$

$$\begin{array}{l|l} 6x^2-5x-14 & x-2 \\ \hline -6x^2+12x & 6x+7 \\ \quad 7x-14 & \\ \quad -7x+14 & \\ \quad (0) & \end{array}$$

$$\begin{array}{l|l} x^3-3x^2-7x+18 & x-2 \\ \hline -x^3+2x^2 & x^2-x-9 \\ \quad -x^2-7x & \\ \quad +x^2-2x & \\ \quad -9x+18 & \\ \quad +9x-18 & \\ \quad (0) & \end{array}$$

$$\lim_{x\to 2} \frac{\cancel{(x-2)}\cdot(6x+7)}{\cancel{(x-2)}\cdot(x^2-x-9)\cdot h(x)} =$$

⊕ Atenção! Cancele! Deixe claro o que fez antes de aplicar o limite!

$$= \frac{6\cdot 2+7}{(2^2-2-9)\cdot h(2)} =$$

$$= \frac{19}{-7\cdot 8} = \frac{19}{-56}$$

$$\lim_{x\to+\infty}\left(\frac{5x-17}{5x+11}\right)^{-\frac{15x}{7}} =$$

$$\lim_{x\to+\infty}\left(\frac{\frac{5x-17}{5x}}{\frac{5x+11}{5x}}\right)^{-\frac{15x}{7}} = \lim_{x\to+\infty}\left(\frac{\frac{5x}{5x}-\frac{17}{5x}}{\frac{5x}{5x}+\frac{11}{5x}}\right)^{-\frac{15x}{7}} =$$

$$= \lim_{x\to+\infty}\left(\frac{1-\frac{17}{5x}}{1+\frac{11}{5x}}\right)^{-\frac{15x}{7}} = \lim_{x\to+\infty}\left[\frac{\left(1+\left(-\frac{17}{5}\right)\cdot\frac{1}{x}\right)^{x}}{\left(1+\frac{11}{5}\cdot\frac{1}{x}\right)^{x}}\right]^{-\frac{15}{7}} =$$

$$= \left(\frac{e^{\left(\frac{-11}{5}\right)}}{e^{\left(\frac{11}{5}\right)}}\right)^{-\frac{15}{7}} = \left(e^{-\frac{11}{5}-\frac{11}{5}}\right)^{-\frac{15}{7}} = \left(e^{-\frac{28}{5}}\right)^{-\frac{15}{7}}$$

$$= e^{\left(-\frac{28}{5} \cdot -\frac{15}{7}\right)} = e^{\left(\frac{-4}{1} \cdot \frac{-3}{1}\right)} = e^{\frac{12}{1}} = e^{12}$$

$$\lim_{x \to 0} \frac{1 - \sec 6x}{x^2} = \lim_{x \to 0} \frac{1 - \frac{1}{\cos 6x}}{x^2} = \lim_{x \to 0} \frac{\frac{\cos 6x}{\cos 6x} - \frac{1}{\cos 6x}}{\frac{x^2}{1}}$$

$$= \lim_{x \to 0} \frac{\frac{\cos 6x - 1}{\cos 6x}}{\frac{x^2}{1}} = \lim_{x \to 0} \frac{(\cos 6x - 1)}{\cos 6x} \cdot \frac{1}{x^2} \cdot \frac{(\cos 6x + 1)}{(\cos 6x + 1)} =$$

$$= \lim_{x \to 0} \frac{(\cos^2 6x - 1)}{x^2 \cdot \cos 6x (\cos 6x + 1)} = \lim_{x \to 0} \frac{-\operatorname{sen}^2 6x}{x^2 \cdot \cos 6x \cdot (\cos 6x + 1)}$$

$$= \lim_{x \to 0} \frac{-\operatorname{sen} 6x}{x} \cdot \frac{\operatorname{sen} 6x}{x} \cdot \frac{1}{\cos 6x \cdot (\cos 6x + 1)} =$$

$$= -6 \cdot 6 \cdot \frac{1}{1 \cdot (1+1)} = \frac{-36}{2} = -\frac{18}{1} = -18$$